

ENGINEERING EXPLORATION

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KEY CONCEPTS



- 1.What is engineering.
- 2.Difference between science and engineering
- 3.Difference between scientist and engineer
- 4.Needs and wants of Computer science engineer in today's world
5. List out various discipline of engineering and map the role of computer science engineer with other branch of engineering to with today's real world product(case study).
6. Misconceptions about computer science engineers
7. Skill required for computer science engineering in 21st century

WHAT IS ENGINEERING?

Engineering is the application of science and mathematics to solve real-world problems by designing, building, improving, and maintaining products, systems, and technologies.

Easy Formula:

Science discovers → Engineer designs → Society benefits



DIFFERENCE BETWEEN SCIENCE AND ENGINEERING

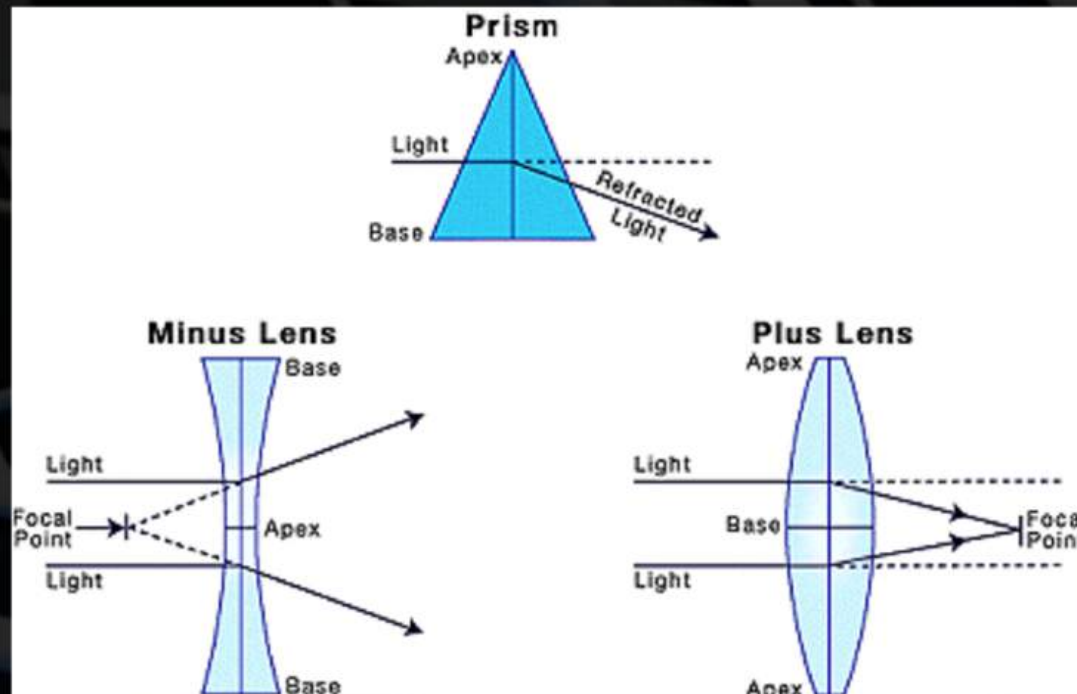
Science:

Science is the systematic study of nature and the universe. It helps us understand why and how things happen by observing, experimenting, and discovering new knowledge. The main goal of science is to explain natural phenomena and develop theories based on evidence.

Engineering:

Engineering is the practical application of scientific knowledge and mathematics to solve real-world problems. Engineers use the principles discovered by scientists to design, build, test, and improve products, systems, and technologies that make human life easier.

EXAMPLE: MANUFACTURING EYEGLASSES

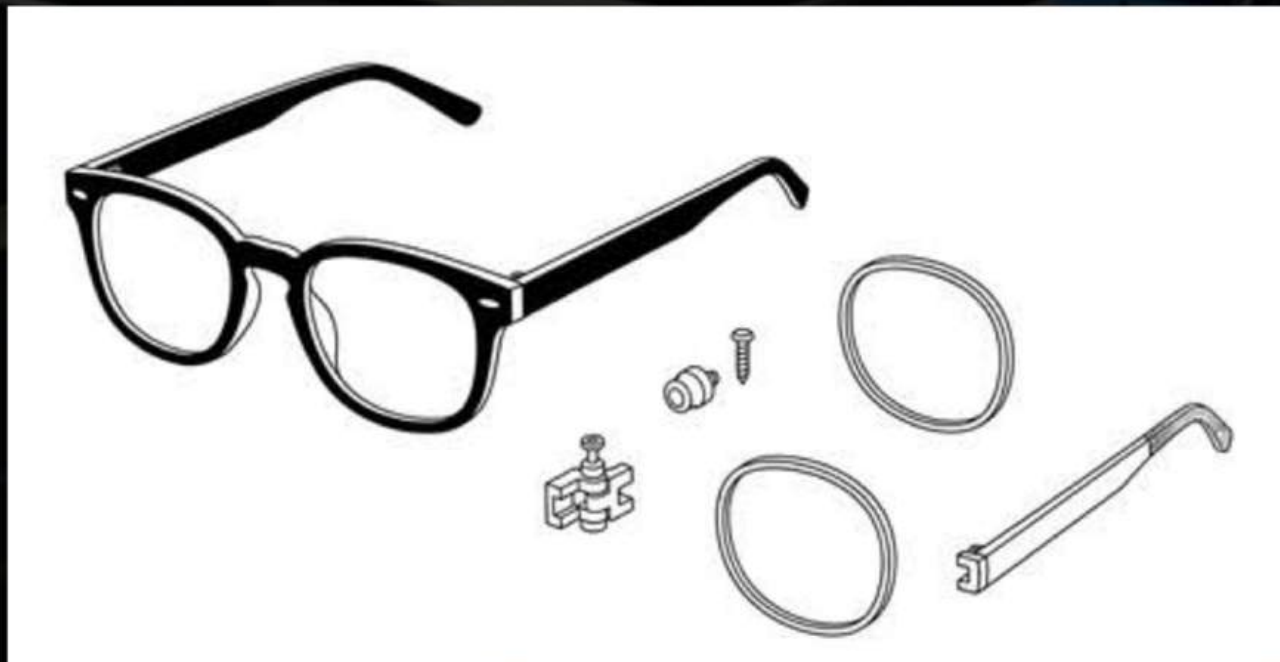


Science

Science reveals that how light bends (refraction) through lenses.

Engineering

Engineers use this knowledge to design and manufacture eyeglasses that help people see clearly.



DIFFERENCE BETWEEN SCIENTIST AND ENGINEER

scientist

A scientist is a person who studies nature and the universe to discover new knowledge. Scientists conduct experiments, make observations, and develop theories to understand why and how things happen.

Main Goal:

- To discover new knowledge.
- To explain natural phenomena.

engineer

An engineer is a person who applies scientific knowledge and mathematics to design, build, and improve products, systems, and technologies that solve real-world problems.

Main Goal:

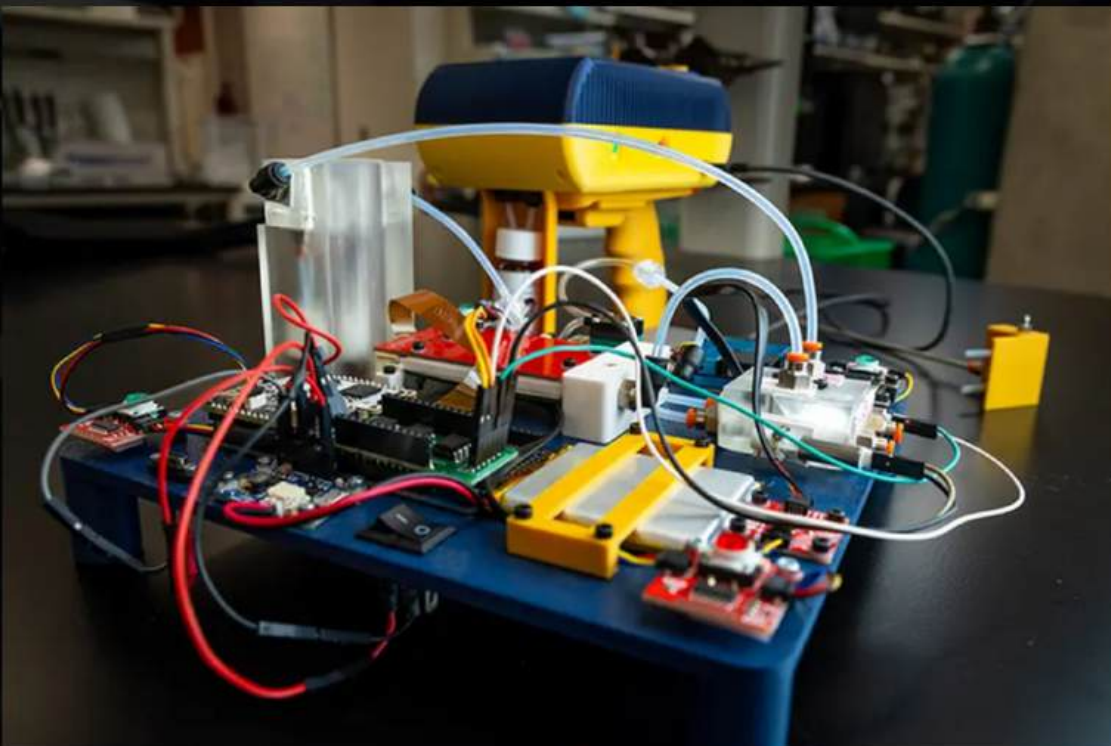
- To solve practical problems.
- To create useful products and technologies.

EXAMPLE :IN VIRUS



○ Scientist

A scientist studies how viruses spread . Scientists collect samples from infected patients and examine them under powerful microscopes and in laboratories.



○ Engineer

An engineer develops software for tracking disease outbreaks or designs medical devices to detect infections.

Various Disciplines of Engineering:

**MECHANICAL
ENGINEERING**

**CIVIL
ENGINEERING**

**ENVIRONMENTAL
ENGINEERING**

**ELECTRICAL
ENGINEERING**

**ELECTRONICS
ENGINEERING**

**BIOMEDICAL
ENGINEERING**

**AEROSPACE
ENGINEERING**

**CHEMICAL
ENGINEERING**

**AGRICULTURAL
ENGINEERING**

**AUTOMOBILE
ENGINEERING**

Misconception about computer science engineer:

www.reallygreatsite.com

1.Misconception:

Computer Science is only about coding.

Reality:

Coding is just one part of Computer Science. It also includes algorithms, artificial intelligence, cybersecurity, data science, cloud computing, databases, networking, and software engineering.

2.Misconception:

Computer Science Engineers only repair computers.

Reality:

Repairing hardware is mainly the job of computer technicians. Computer Science Engineers design, develop, test, and maintain software and computing systems.

3.Misconception:

You must be a mathematics genius to study CSE..

Reality:

Basic mathematics and logical thinking are important, but persistence and problem-solving skills matter more than being a math genius.

4.Misconception:

Computer Science Engineers work only in IT companies.

Reality:

CSE graduates work in many industries, including healthcare, banking, education, manufacturing, agriculture, aerospace, entertainment, and government.

5. Misconception:

Artificial Intelligence will replace all programmers.

Reality:

AI helps automate repetitive tasks, but engineers are still needed to design systems, solve complex problems, ensure security, and make technical decisions.

SKILLS REQUIRED FOR COMPUTER SCIENCE ENGINEERING IN THE 21ST CENTURY



Technical Skills

- Programming (Python, Java, C++, JavaScript)
- Data Structures and Algorithms
- Database Management (SQL, NoSQL)
- Web and Mobile Application Development



Problem-Solving Skills

- Analyze complex problems
- Develop efficient and innovative solutions
- Apply logical and critical thinking



Communication and Teamwork

- Communicate ideas clearly
- Collaborate effectively with multidisciplinary teams
- Write technical documentation and reports



Adaptability and Continuous Learning

- Learn new programming languages and tools
- Keep up with emerging technologies
- Adapt to changing industry requirements



Professional and Ethical Skills

- Time and project management
- Professional ethics and responsibility
- Awareness of data privacy and cybersecurity practices

ROLE OF COMPUTER SCIENCE ENGINEER WITH REAL-WORLD PRODUCTS (CASE STUDY)

NEXT SLIDE



Mechanical Engineering:

Role of Computer Science Engineer: Develops embedded software, automation systems, and control algorithms for smart machines.

Real-World Product: Self-driving cars and industrial robots.



Civil Engineering:

Role of Computer Science Engineer: Develops software for smart infrastructure, traffic management, and structural health monitoring.

Real-World Product: Smart city traffic management systems.



Electrical Engineering:

Role of Computer Science Engineer: Designs software for power monitoring, smart grids, and energy management systems.

Real-World Product: Smart electricity meters.



Electronics and Communication Engineering:

Role of Computer Science Engineer: Develops firmware, communication software, and Internet of Things (IoT) applications.

Real-World Product: Smartphones and smart home devices.



Aerospace Engineering:

Role of Computer Science Engineer: Develops flight control software, navigation systems, and simulation applications.

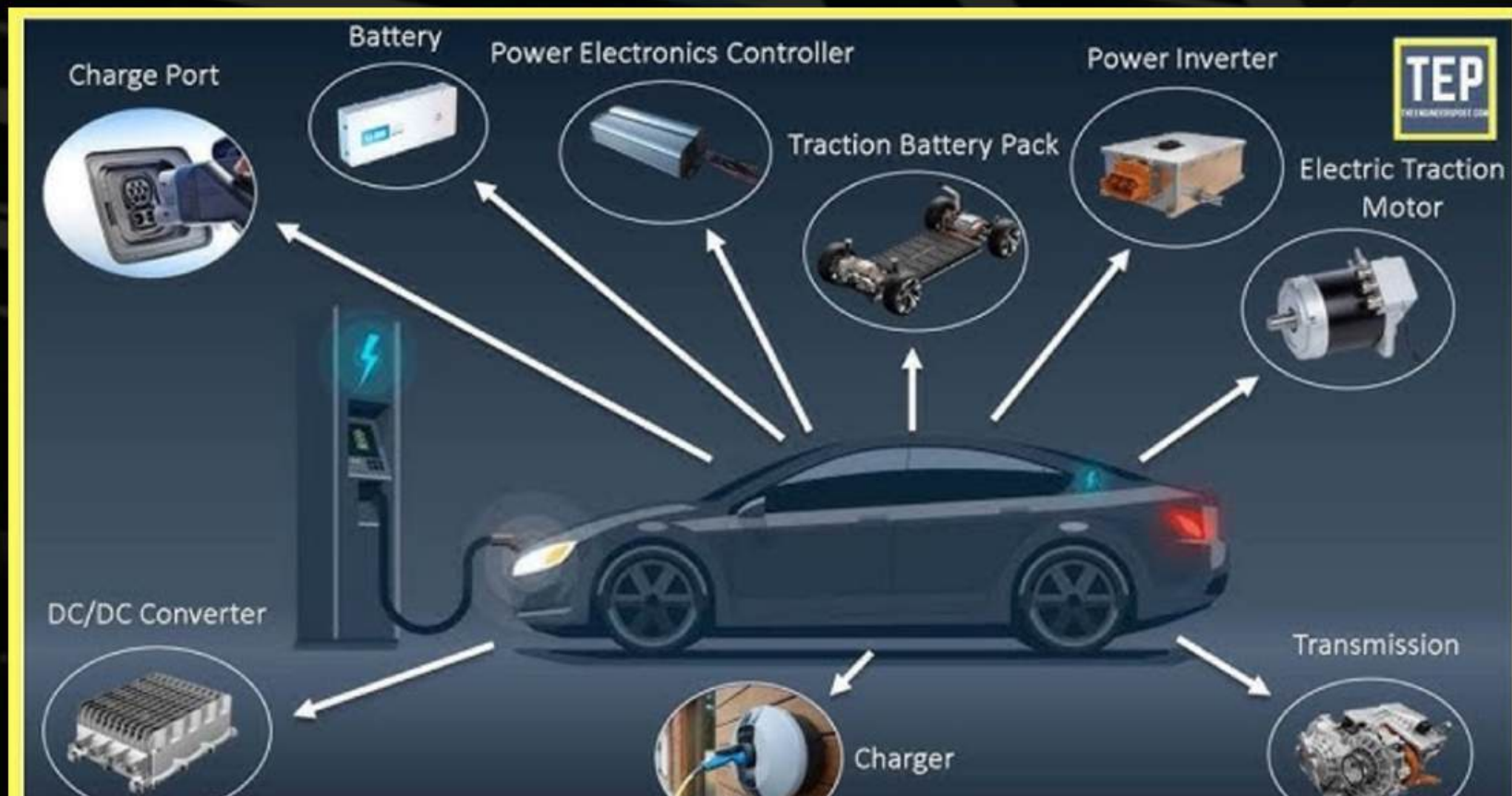
Real-World Product: Aircraft autopilot systems and drones.

CASE STUDY: SMART EV

NEXT SLIDE

- **Mechanical Engineering:** Designs the vehicle body, chassis, and mechanical components.
- **Electrical Engineering:** Develops the battery, motor, and power distribution system.
- **Electronics and Communication Engineering:** Integrates sensors, controllers, and communication modules.
- **Computer Science Engineering:** Develops autonomous driving software, battery management systems, navigation, mobile applications, cybersecurity, and cloud connectivity.
- **Industrial Engineering:** Optimizes manufacturing and production processes

FINAL PRODUCT:



Smart Electric Vehicle (EV)

The Smart EV is a real-world example of multidisciplinary engineering, where the Computer Science Engineer acts as the backbone of the intelligent software system, enabling features such as navigation, autonomous driving, battery monitoring, mobile app connectivity, over-the-air software updates, and vehicle cybersecurity.



THANK YOU!